# Venkata Lakshmi Parimala Pasupuleti

venkatalakshmiparimalap@gmail.com | +1-703-261-5410 | Fairfax, VA | LinkedIn | GitHub

## SUMMARY

Backend software engineer with 6+ years of experience building scalable APIs, data pipelines, and multi-tenant SaaS platforms for housing. Spent 6 years at Entrata building property management infrastructure across affordable housing, conventional apartments, and student housing communities -- PostgreSQL-heavy backend, async processing, role-based access control, PHI encryption, and CI/CD. Currently at Intuit building Python/FastAPI services on AWS. Strong in database design, query optimization, REST APIs, and backend security. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

## CORE COMPETENCIES

PostgreSQL | REST APIs | Python | Supabase | Serverless Functions | Role-Based Access Control | Data Encryption | Real-Time Data Sync | AWS (Lambda, S3, RDS, SQS) | CI/CD (GitHub Actions, Vercel) | Docker | Microservices | Database Design | Query Optimization | FastAPI | Node.js / TypeScript | React | RabbitMQ | Code Review / Mentoring

## EXPERIENCE

### Intuit

Jun 2026 -- Present

**Software Engineer** -- Virginia, USA (Hybrid)

- Build and deploy Python/FastAPI backend services on AWS (Lambda, S3, RDS, API Gateway, SQS) with GitHub Actions CI/CD and Docker containerization.
- Design REST API contracts with structured-output schemas and validation, ensuring downstream services receive consistent, traceable responses.
- Build ML pipelines (scikit-learn, XGBoost) from data ingestion and feature engineering through model evaluation and prediction APIs.
- Implement input validation, sensitive-data handling, and role-based access patterns for financial data services.
- Instrument services with Prometheus and OpenTelemetry for production observability; conduct code reviews and mentor junior engineers.

### Entrata -- Apartment, Affordable Housing & Student Housing SaaS

Oct 2020 -- Jul 2024

**Senior Software Engineer** -- Multi-tenant platform, 100K+ monthly active users

- Architected and maintained **PostgreSQL-heavy backend** across affordable housing, conventional apartment, and student housing communities -- complex relational schemas covering lease management, eligibility, rent calculations, and occupancy tracking.
- Built **role-based access control** and multi-tenant data isolation for property managers, leasing staff, and residents across thousands of communities including student housing portfolios.
- Implemented **Libsodium-encrypted data pipelines** covering 100% of resident PHI (SSNs, phone numbers) to meet institutional data protection standards; zero security incidents.
- Developed **RabbitMQ-based real-time async processing** for rent and eligibility updates across thousands of tenant records, with idempotency guarantees and retry logic.
- Optimized 200+ SQL queries using execution plan analysis and indexing -- 30% average performance improvement on a complex multi-tenant schema.
- Led AWS S3 migration for document storage (lease agreements, certification forms); created 56 state-specific housing documents via automated merge-field pipelines.
- Delivered 20+ features end-to-end across 120+ Agile sprints; reviewed 300+ pull requests and mentored junior engineers.

### Entrata India

Sep 2018 -- Oct 2020

**Junior Software Engineer** -- Pune, India

- Built property management modules (Python, PHP, Java, PostgreSQL) for affordable housing and student housing rental management workflows.
- Delivered large-scale data import and reconciliation workflows processing 1.6M HUD/Tax Credit records with validation and rollback checks.

- Led AWS S3 cloud migration from legacy server storage; automated 56 state-specific housing certification documents.

## PROJECTS

### Leasel -- Multi-Agent Housing Compliance Platform

Python · LangGraph · FastAPI · PostgreSQL · CI/CD

Backend platform for housing compliance operations -- lease document ingestion, OCR extraction, section-aware chunking, and a REST API serving compliance queries with cited answers. Built role-based access control so eligibility and financial decisions always route to staff rather than being resolved by the system. LLM-as-judge CI harness across 1,240 test cases: 93.6% precision, 87.9% F1.

github.com/venkatalakshmiparimala/Lease_complaince_rag

### AI on FHIR -- Clinical NLP Backend

Python · Flask · PostgreSQL · REST APIs · RBAC · Docker

Backend API translating English/Spanish clinical questions into FHIR REST API calls using NLP and LLMs. Implemented role-based access control for HIPAA-style data isolation across user tiers. Containerized with Docker; reduced manual clinical lookup time by ~60%.

github.com/venkatalakshmiparimala/AI_on_FHIR

### AgentProbe -- LLM Evaluation Toolkit

Python · FastAPI · Serverless · GitHub Actions CI/CD

15-endpoint FastAPI backend with serverless-compatible architecture routing prompts across five LLM providers. CI regression tests with golden datasets for continuous quality assurance.

github.com/venkatalakshmiparimala/AgentProbe

## EDUCATION

**M.S. Data Analytics Engineering** — George Mason University, Virginia — May 2026 | GPA 3.87

**B.Tech Computer Science & Engineering** — JNTU Anantapur — 2017